CYTECH ACADEMY

Cybersecurity Foundations

Complete Study Notes - beginner friendly, exam and lab ready.

(c) CyTech. For enrolled students. Unauthorized attack is illegal - practice only in labs.

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What is Cybersecurity?

Cybersecurity is the practice of protecting systems, networks, programs and data from digital attacks. These attacks usually aim to access, change or destroy sensitive information, extort money, or interrupt normal business processes.

The discipline spans people, process and technology: even the strongest encryption fails if a user reuses a weak password. A professional therefore thinks in layers - technical controls, policies, and continuous education.

The CIA Triad

Confidentiality: only authorized parties can read data (encryption, access control, least privilege). Integrity: data is accurate and unmodified except by authorized action (hashing, digital signatures, checksums). Availability: systems and data are accessible when needed (redundancy, backups, DDoS protection).

Every security decision maps to one or more of these three pillars. When you analyze an incident, ask which pillar was broken; when you design a control, ask which pillar it protects.

Threat Actors and Attack Types

Actors range from script kiddies and cybercriminals (ransomware, fraud) to hacktivists, insider threats and nation-state APT groups. Motivations: money, espionage, ideology, disruption, reputation.

Common attacks: phishing and social engineering, malware (viruses, worms, trojans, ransomware, spyware), credential attacks (brute force, stuffing, spraying), MITM, SQL injection, XSS, zero-days and supply-chain compromise.

Core Defenses

Defense in depth: firewalls and network segmentation, EDR/antivirus, patch management, MFA, strong password policy and managers, encrypted transport (TLS) and storage (AES-256), secure configuration baselines, logging and SIEM monitoring, regular backups with tested restore.

Security awareness training is the highest-ROI control: most breaches begin with a human click. Simulate phishing, teach verification habits, and make reporting easy.

Careers and Certifications

Roles: SOC analyst, penetration tester, security engineer, DFIR analyst, GRC analyst, malware reverse engineer, CISO. Entry certs: CompTIA Security+, Google Cybersecurity Certificate. Practical: eJPT, PNPT. Advanced: OSCP, CEH (theory), GCIH, CHFI.

Build a home lab (VirtualBox + Kali + a vulnerable VM like Metasploitable or DVWA), document everything you learn publicly, and practice on legal platforms: TryHackMe, HackTheBox,

PicoCTF.